



KunHo Heo

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Research Interests

3D Vision: Scene Understanding

- Scene Graph, Affordance Grounding

3D Vision: Character Animation

- Human Motion Generation, Human-Object Interaction

2D Vision: Perception

- Few-Shot Object Detection

Education

Kyung Hee University

Mar. 2025 – Feb. 2027 (Expected)

M.S. in Software Convergence

- GPA: 4.45 / 4.5

Kyung Hee University

Mar. 2021 – Feb. 2025

B.S. in Software Convergence

- GPA: 4.04 / 4.5

Research Experience

Visual Science Lab [🔗](#), Kyung Hee University

Mar. 2025 – Present

M.S. Student (**Lab Manager**)

- Advisor: Prof. MyeongAh Cho

Visual Science Lab [🔗](#), Kyung Hee University

Sep. 2024 – Feb. 2025

Undergraduate Research Intern

- Advisor: Prof. MyeongAh Cho

*IIIXR Lab [🔗](#), Kyung Hee University **relocated to Korea University*

Jan. 2023 – Jun. 2024

Undergraduate Research Intern

- Advisor: Prof. HyeongYeop Kang

Publications

International Conference

Rethinking Prototype-based Similarity Learning for Few-Shot Object Detection

KunHo Heo^{*}, Seungjae Kim^{*}, Wongyu Lee^{*}, SuYeon Kim, MyeongAh Cho **Equal Contribution*

Under Review

ParTY: Part-Guidance for Expressive Text-to-Motion Synthesis

[Paper](#) [🔗](#)

KunHo Heo, SuYeon Kim, Yonghyun Gwon, Youngbin Kim, MyeongAh Cho

IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)

Object-Centric Representation Learning for Enhanced 3D Semantic Scene Graph Prediction

Paper [↗](#)

KunHo Heo^{*}, GiHyun Kim^{*}, SuYeon Kim, MyeongAh Cho ^{*}*Equal Contribution*

The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)

International Journal

Revisiting Object-Centric Learning for Long-Tailed 3D Semantic Scene Graph Prediction

GiHyun Kim, **KunHo Heo**, SuYeon Kim, MyeongAh Cho

Under Review (IEEE Transactions on Image Processing)

Domestic Conference

Word-Motion Cross Attention for Fine-Grained Text Guided Human-Human Interaction Motion Generation

Yonghyun Gwon, **KunHo Heo**, Youngkwan Cho, SuYeon Kim, Hayoung Lee, MyeongAh Cho

Korea Software Congress (KSC), 2025

Distance-based Bi-directional Edge Update Graph Neural Network for 3D Scene Graph Generation

SuYeon Kim, GiHyun Kim, **KunHo Heo**, MyeongAh Cho

Korea Computer Congress (KCC), 2025

Enhancing Text-driven Human Motion Generation via Part Transfer and Sequential Attention Learning

KunHo Heo, Seung Joon Lee, Hayoung Lee, MyeongAh Cho

Korea Software Congress (KSC), 2024

Body-Part Embedding for Text-driven Human Motion Generation

KunHo Heo, HyeongYeop Kang

Korea Computer Congress (KCC), 2024

Personality labeling methods with motion parameters

KunHo Heo, HyeongYeop Kang

Korea Software Congress (KSC), 2023

Projects

Development of AGI platform for multisensory social robot

Institute for Information communication
Technology Planning and Evaluation (IITP)
Apr. 2025 - Dec. 2025

- **Project Manager**
- Developed the **Vision system** for multisensory perception
- Researched **Few-Shot Object Detection** to enhance adaptability to dynamic environments under limited data

Development of AI technology capable of robustly recognizing abnormal and dangerous situations and behaviors during night and bad weather conditions

Korea Planning & Evaluation Institute
of Industrial Technology (KEIT)
Jul. 2024 - Mar. 2028

- **Image restoration** and **Data pre-processing** under adverse visual conditions

Anomaly pre-detection and response for intelligent surveillance system: high-level interpretation and response based on the visual-language fusion

National Research Foundation
of Korea (NRF)
Sep. 2024 - Aug. 2027

- **Vision-language model integration** for surveillance anomaly interpretation and response

Patents

Domestic Patent

Method and Apparatus for Generating Whole-Body Motion by Synthesizing Part-wise Motions Dec. 2025

- KR-Application No. 10-2025-0200238

Method and Apparatus for Generating 3D Scene Graphs Dec. 2025

- KR-Application No. 10-2025-0188361

Software Copyright

Improved Sound Source Localization Model via Audio Signal Enhancement Sep. 2025

- KR-Registration No. C-2025-035472

Honors & Awards

Outstanding Researcher Scholarship 2025, 2026

- Kyung Hee University

Academic Scholarship 2021 – 2024

- Kyung Hee University

Grand Prize Dec. 2024

- Undergraduate Paper Competition, Korea Software Congress (**KSC**)

Grand Prize (Overall 1st Prize across all categories) Nov. 2024

- SW Festival, Kyung Hee University

1st Prize in Immersive Media Category Nov. 2024

- SW Festival, Kyung Hee University

National Research Foundation (NRF) President's Award Nov. 2024

- Immersive Media Competition, Convergence and Open Sharing System (**COSS**)

Grand Prize May. 2024

- AI&SW Hackathon, Kyung Hee University

1st Prize in Immersive Media Category Nov. 2023

- SW Festival, Kyung Hee University

Teaching Experience

Teaching Assistant Spring 2025, Spring 2026

- Artificial Intelligence and Game Programming

Teaching Assistant Fall 2025

- Game Engineering
- Open Source Software Development Methods and Tools
- Design Thinking